

Hw 7

1.

(a)

poset A: the diagram is the chain $c \prec b \prec a$ so this means

$$R_A = \{(a, a), (b, b), (c, c), (c, b), (b, a), (c, a)\}.$$

poset B this is $(\mathcal{P}(\{a, b\}), \subseteq)$ on the four elements $\emptyset, \{a\}, \{b\}, \{a, b\}$ so then

$$\begin{aligned} R_B = \{ & (\emptyset, \emptyset), (\{a\}, \{a\}), (\{b\}, \{b\}), (\{a, b\}, \{a, b\}), \\ & (\emptyset, \{a\}), (\emptyset, \{b\}), (\emptyset, \{a, b\}), \\ & (\{a\}, \{a, b\}), (\{b\}, \{a, b\}) \}. \end{aligned}$$

poset C: the cover relations are $1 \prec 3$, $1 \prec 4$, and $2 \prec 4$, so

$$R_C = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (1, 4), (2, 4)\}.$$

(b)

poset A:

- Minimal elements: $\{c\}$
- Maximal elements: $\{a\}$
- Least element: c
- Greatest element: a

Poset (B):

- Minimal elements: $\{\emptyset\}$
- Maximal elements: $\{\{a, b\}\}$
- Least element: $\emptyset$
- Greatest element: $\{a, b\}$

poset (C):

- Minimal elements: $\{1, 2\}$
- Maximal elements: $\{3, 4\}$
- Least element: does not exist
- Greatest element: does not exist

(c)

1. for 1 and 2:

$$\text{GLB}(1, 2) \text{ does not exist,} \quad \text{LUB}(1, 2) = 4.$$

2. for 3 and 4:

$$\text{GLB}(3, 4) = 1, \quad \text{LUB}(3, 4) \text{ does not exist.}$$

3. for 1 and 4:

$$\text{GLB}(1, 4) = 1, \quad \text{LUB}(1, 4) = 4.$$

2.

- (A) is a lattice
- (B) is a lattice
- (C) is not a lattice and for the pair 5, 6 there's no GLB because there is no common lower bound of 5 and 6
- (D) is a lattice
- (E) isn't a lattice and for the pair 4, 5 there is no LUB because the common upper bounds are 2, 3, 1 and the least one does not exist since 2 and 3 are incomparable minimal upper bounds
- (F) is a lattice
- (G) isn't a lattice and for the pair 4, 5 there is no LUB because the common upper bounds are 2, 3, 1 and the least one doesn't exist since 2 and 3 are incomparable minimal upper bounds

3.

(a) $(S, \preceq)$ where $S = \{1, 2, 3, 5, 6, 10, 15, 30\}$ and $x \preceq y \iff x \mid y$

i. the cover relations are

$$\begin{aligned} 1 < 2, 1 < 3, 1 < 5, \\ 2 < 6, 2 < 10, 3 < 6, 3 < 15, 5 < 10, 5 < 15, \\ 6 < 30, 10 < 30, 15 < 30. \end{aligned}$$

so the Hasse diagram has 1 at the bottom and 2, 3, 5 above it then 6, 10, 15 and 30 at the top.

ii. yes, for any two elements of S both their greatest common divisor and least common multiple are also in S so every pair has both a meet and a join

iii. in this poset we got

$$x \wedge y = \gcd(x, y), \quad x \vee y = \text{lcm}(x, y).$$

examples:

$$\begin{aligned} 6 \wedge 10 &= 2, & 6 \vee 10 &= 30. \\ 10 \wedge 15 &= 5, & 10 \vee 15 &= 30. \end{aligned}$$

(b) $(S, \preceq)$ **where** $S = \mathcal{P}(\{1, 2, 3\})$ **and** $x \preceq y \iff x \subseteq y$

i. the elements are

$$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$$

the hasse diagram is the boolean lattice B_3 : $\emptyset$ at the bottom and the three 1-element subsets above it so the three 2element subsets above those and $\{1, 2, 3\}$ at the top

ii. yes it's a lattice because every pair of subsets has both an intersection and a union so every pair has both a meet and a join

iii in this poset we have

$$A \wedge B = A \cap B, \quad A \vee B = A \cup B.$$

examples:

$$\begin{aligned} \{1, 2\} \wedge \{2, 3\} &= \{2\}, & \{1, 2\} \vee \{2, 3\} &= \{1, 2, 3\}. \\ \{1\} \wedge \{1, 3\} &= \{1\}, & \{1\} \vee \{1, 3\} &= \{1, 3\}. \end{aligned}$$